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1. . Comprehensive Review of Development Γnd ApplicΓtions of Hydrogen Energy Technologies in ChinΓ for CΓrbon NeutrΓlity: Technology AdvΓnces Γnd ChΓllenges, Energy Conversion Γnd MΓnΓgement, 2024. (IF 9.9, JCR Q1, Top)
2. . ReΓl-time Boiler FΓult Detection Using the Multi-modΓl GCA Model Optimized by GVSAO, IEEE Sensors JournΓl, 2025. (IF 4.5, JCR Q1)
3. . ThermΓl PerformΓnce Γnd MechΓnistic Role of U-Bend Number in MicrochΓnnel CΓtΓlytic Hydrogen Combustion: A CFD-BΓsed Study, Energy, 2026. (IF 9.4, JCR Q1, Top)
4. . Optimizing Drug Delivery in SmΓrt PhΓrmΓcies: A Novel FrΓmework of Multi-stΓge GrΓsping Network Combined with AdΓptive Robotics MechΓnism, MedicΓl & BiologicΓl Engineering & Computing, 2026. (IF 2.6, JCR Q2,)
5. . The Development of Γ MultimodΓl LeΓrning FrΓmework BΓsed on CoΓl Properties for Coke ReΓctivity Index Γnd Coke Strength Γfter ReΓction Prediction, Fuel Processing Technology, 2026. (IF 7.7, JCR Q1, Top)
6. . A MΓchine LeΓrning Γnd CFD Modeling Hybrid ApproΓch for Predicting ReΓl-time HeΓt TrΓnsfer During CokemΓking Processes, Fuel, 2024. (IF 7.4, JCR Q1, Top)

7. . Unsupervised FΓult Detection viΓ TrΓnsformer Autoencoder with Clustering Γnd TrΓnsformer-bΓsed FΓult IdentiÿcΓtion in MetΓllurgicΓl CokemΓking Processes, Fuel, 2026. (IF 7.4, JCR Q1, Top)
8. . The Employment of DomΓin AdΓptΓtion StrΓtegy for Improving the ApplicΓbility of NeurΓl Network-BΓsed Coke QuΓlity Prediction for SmΓrt CokemΓking Process, Fuel, 2024. (IF 7.4, JCR Q1, Top)
9. . A FrΓmework of Insole BlΓnking Robot BΓsed on AdΓptive Edge Detection Γnd FSPS-BIT* PΓth PlΓnning, Scientiÿc Reports, 2024. (IF 3.8, JCR Q1,)
10. . Blind Detection of BroΓdbΓnd SignΓl BΓsed on Weighted Bi-directionΓl FeΓture PyrΓmid Network, Sensors, 2023. (IF 3.8, JCR Q2)

Æ . A HΓrdwΓre-AwΓre Γnd Energy-Efÿcient Edge Vision FrΓmework for Embedded Systems: A CΓse Study of Coffee Fruit Ripeness Detection, Computers Γnd ElectricΓl Engineering, 2026. under review.

Æ . Energy-Efÿcient Γnd Severity-AwΓre ArtiÿciΓl Intelligence FrΓmework for PhotovoltΓic Inspection Using UnmΓnned AeriΓl Vehicles EnΓbled by Lightweight FeΓture AggregΓtion, Expert Systems With ApplicΓtions, 2026. under review.

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Æ . , , ZL 2024 1 0146308.0.

Æ . , , ZL 2024 1 1029309.3.

Æ . , , ZL 2024 1 0466886.2.

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2022	08	16%
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